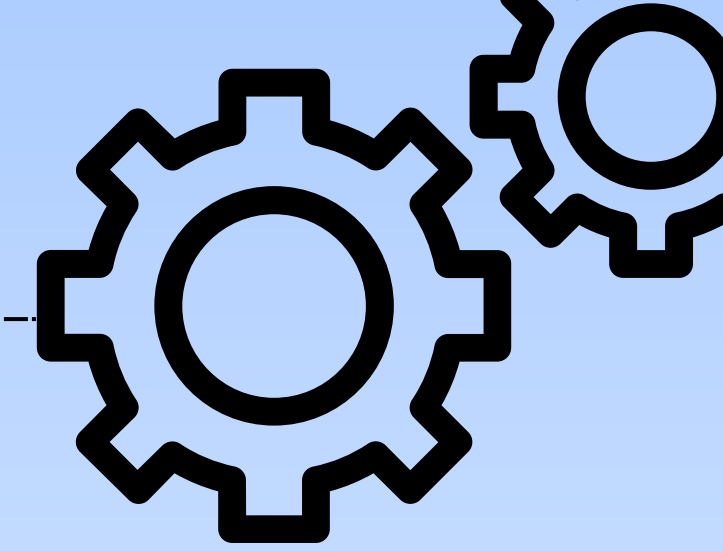
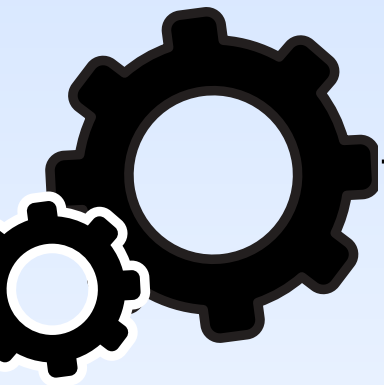




DESIGN PORTFOLIO



GANESHAN GOWTHAN | B.Eng. (Hons.) Mech. Eng.

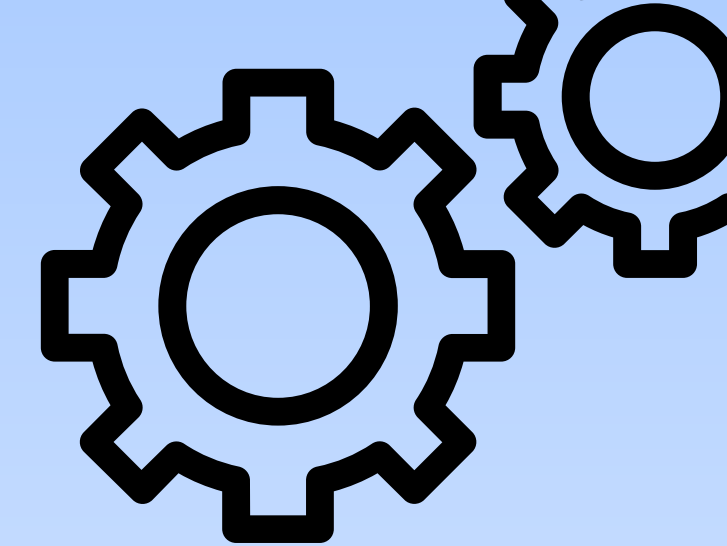
MECHANICAL | MECHATRONICS ENGINEER

 ganeshangowthan1998@gmail.com

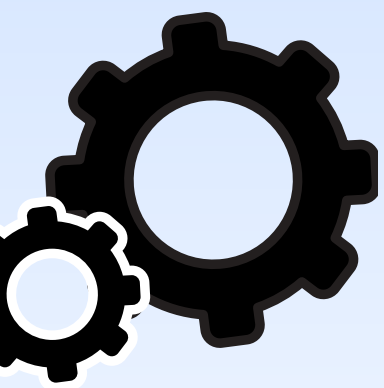
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PROFESSIONAL PROJECTS



STRUCTURAL OPTIMIZATION AND MANUFACTURING OF AN ELECTRIC MOTORCYCLE FRAME

THERMALR INDUSTRIES (PVT) LTD.

Engineering Problems:

1. Excessive wheelbase reduced maneuverability and cornering agility
2. High seat height negatively affected rider ergonomics and accessibility
3. Asymmetrical frame geometry caused uneven load distribution and inconsistent handling
4. Frame structure contained unnecessary material, increasing overall vehicle weight
5. Existing design lacked optimization for manufacturability and assembly

Objectives:

1. Re-design the motorcycle to eliminate the engineering problems
2. Ensure a safe design during the worse case loading conditions

Design and Analysis Tools:

1. SolidWorks - Weldment Design, Assembly
2. ANSYS - Load Analysis and Structural Optimization

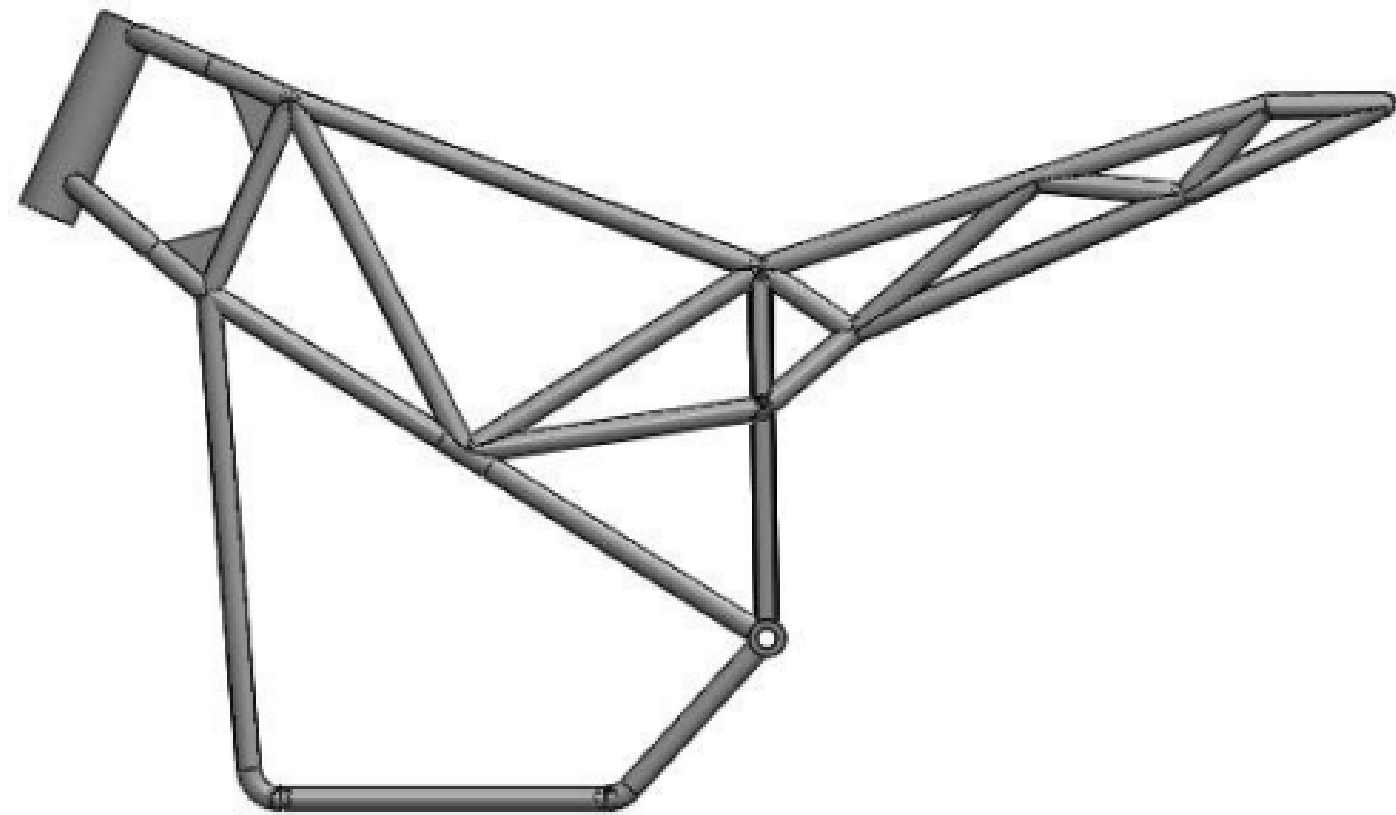


Figure 1: Previous Frame Design - 18.5kg



Figure 2: Final Frame Design - 12.6kg

Design Iterations

Material - Q195 Mild Steel

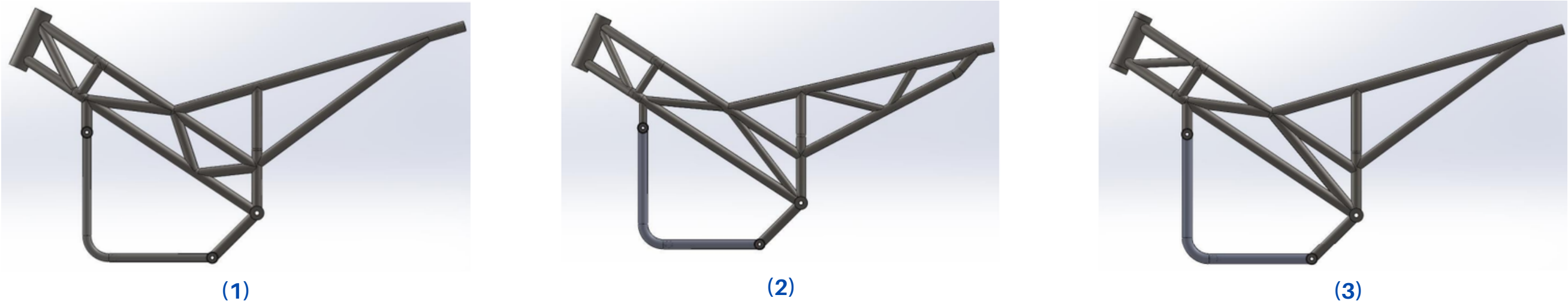


Figure 3: Design Iterations

Structural Analysis:

- 1. Torsional Loading
- 2. Lateral Loading

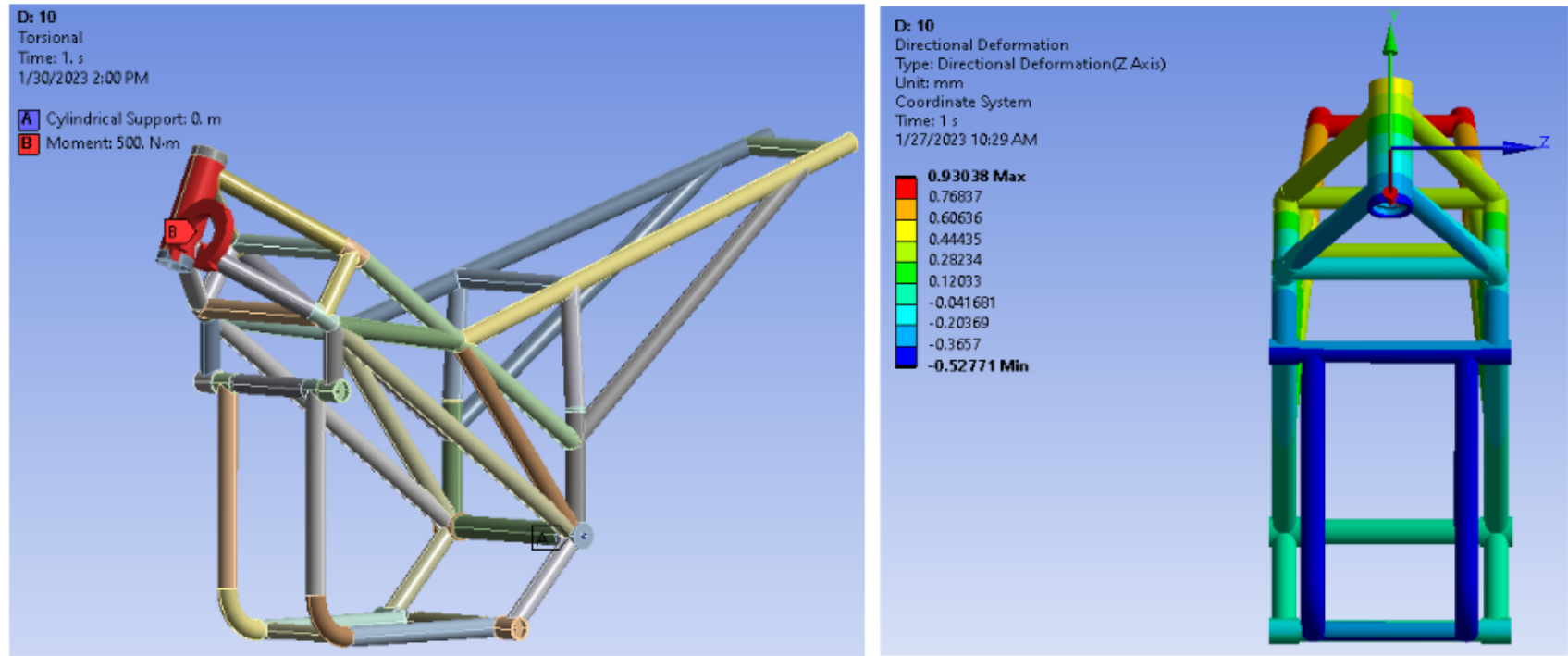


Figure 4: Structural Analysis

Results:

Iteration	Torque (Nm)	Lateral force (N)	$K_t max (kNm/^\circ)$	$K_l max(kN/m)$
10	500	1000	1.2	0.8
11	500	1000	1.4	0.8
12	500	1000	1.3	0.7

- According to Vittore Cossalter’s book of Motorcycle Dynamics, the range of values of torsional and lateral stiffnesses for 1000 cc motorcycles are $3 - 7kNm/^\circ$ and $1 - 3 kN/m$ with the engine fitted.
- Since the motorcycle designed by ThermalR Industries doesn’t consist of an engine and is less powerful compared to a 1000 cc motorcycle, torsional stiffness values between $1 - 2kNm/^\circ$ are acceptable.

Conclusion:

Decided to proceed with Iteration (3) according to the results and ease of manufacture.

PROOF-OF-CONCEPT DESIGN AND FABRICATION OF A MOTORCYCLE FRAME WELDING JIG

THERMALR INDUSTRIES (PVT) LTD.

Overview:

This project focused on the design and manufacturing of a custom welding jig specifically for assembling a trellis frame for an electric motorcycle. The goal was to enhance precision, efficiency, and repeatability during the welding process.

Objectives:

1. Ensure proper alignment of the frame's tube profiles during welding
2. Facilitate easy access for welding without compromising stability.

Design Tools:

SolidWorks - Assembly, Weldments and Sheet metal

Material Selection:

Q195 Mild Steel Box Bars and 1mm thick Sheet metal

Manufacturing Methods:

MIG Welding, Plasma Notching, Plasma Cutting, Turning, Milling and Sheet Metal Bending

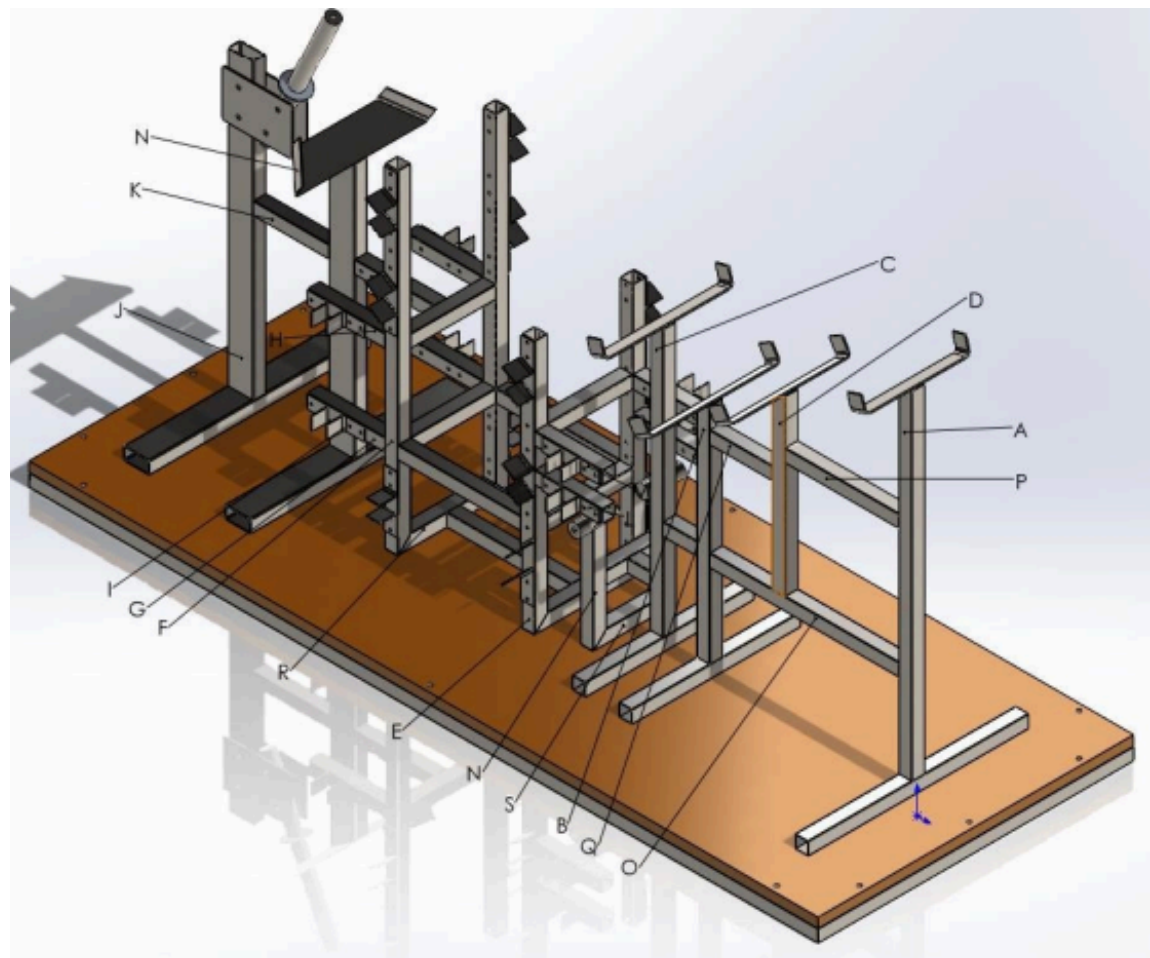


Figure 5: Final Design of the Welding jig

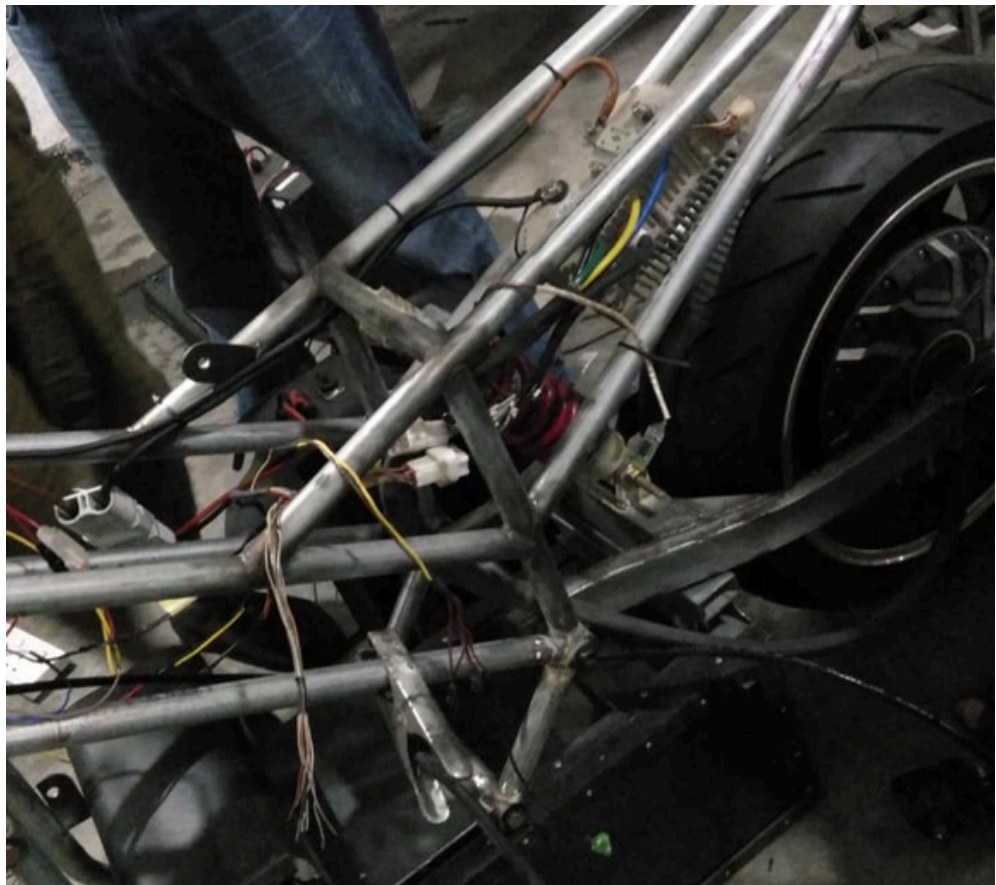


Figure 6: Manufactured Motorcycle Frame

Conclusion:

Enhanced the overall quality of the motorcycle frame with a few defects and rework

DESIGN OF BODY PANELS FOR THE ELECTRIC MOTORCYCLE FRAME ASSEMBLY

THERMALR INDUSTRIES (PVT) LTD.

Overview:

This project focused on the design of the body panels for the first version of the electric motorcycle frame. This was the first job that was assigned to me when I joined ThermalR industries (Pvt) Ltd.

Objectives:

1. Create visually appealing and aerodynamic body panels
2. Facilitate easy assembly and disassembly

Design Tools:

1. SolidWorks - 2D and 3D Sketching, Projected Curves and Surface Modelling
2. SolidWorks Visualize for Rendering and Aesthetics

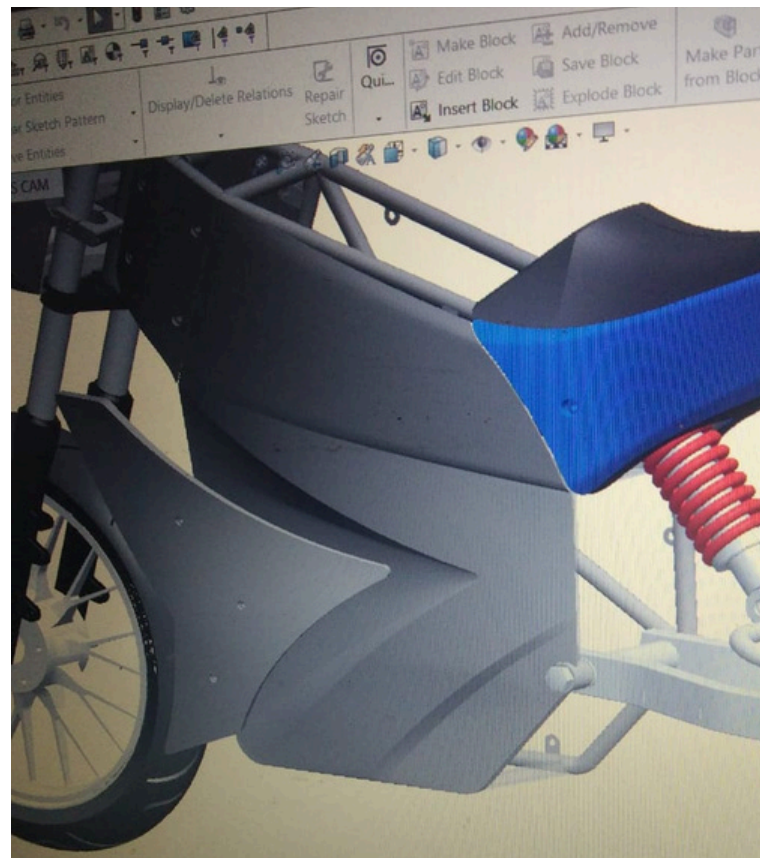


Figure 7: Design and Assembly

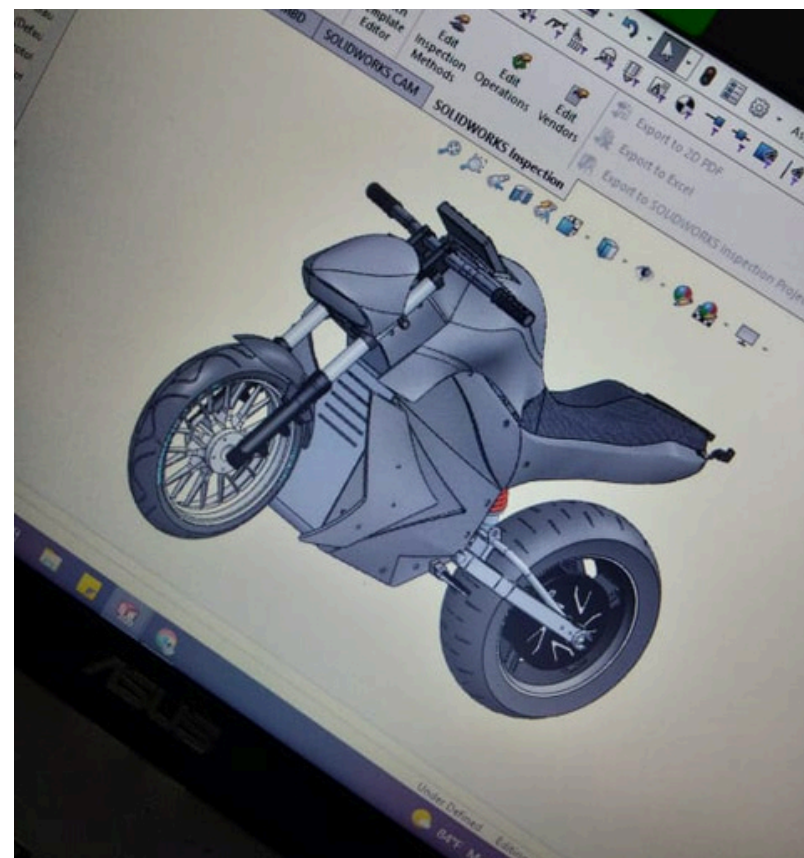


Figure 8: Finalized Model



Figure 9: Rendered Model

DESIGN, PROTOTYPING AND TESTING OF A CUSTOM SHUTTLECOCK DISPENSER MECHANISM

LANKATRONICS (PVT) LTD.

Overview:

This project involved the design, prototyping, and testing of a custom shuttlecock dispenser system efficiently to efficiently manage shuttlecock dispensing with three operational states: Feed, Hold, and Release.

Objectives:

1. Incorporate two Iris Shutter mechanisms to control the dispensing using NEMA17 Stepper Motors
2. Ensure the system is compact and portable

Design Tools:

SolidWorks - Solid Modelling and Motion Study

Prototyping:

3D Printing using PLA+

Testing:

1. Prototyped a breadboard control circuit using two A4988 Stepper Motor Drivers to control the NEMA17 Stepper Motors.
2. Programmed the system using Arduino IDE and tested for functionality,

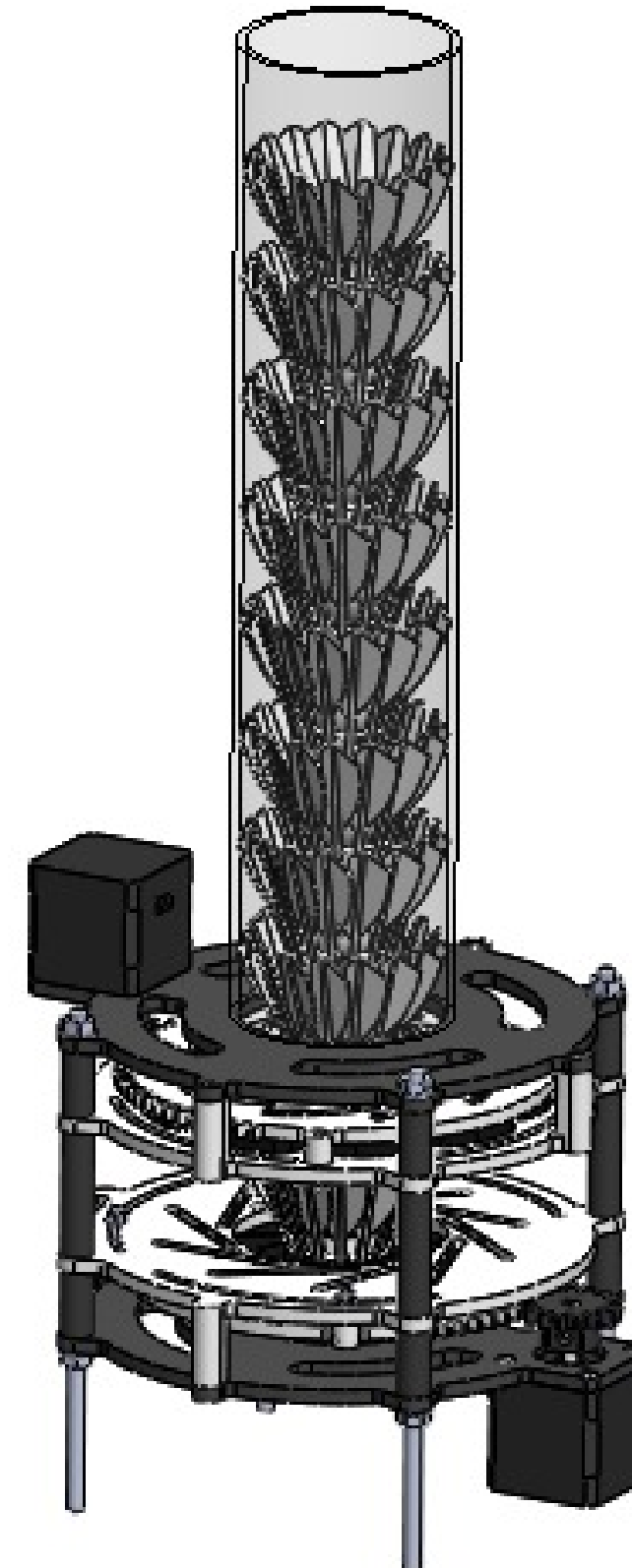


Figure 10: 3D Model

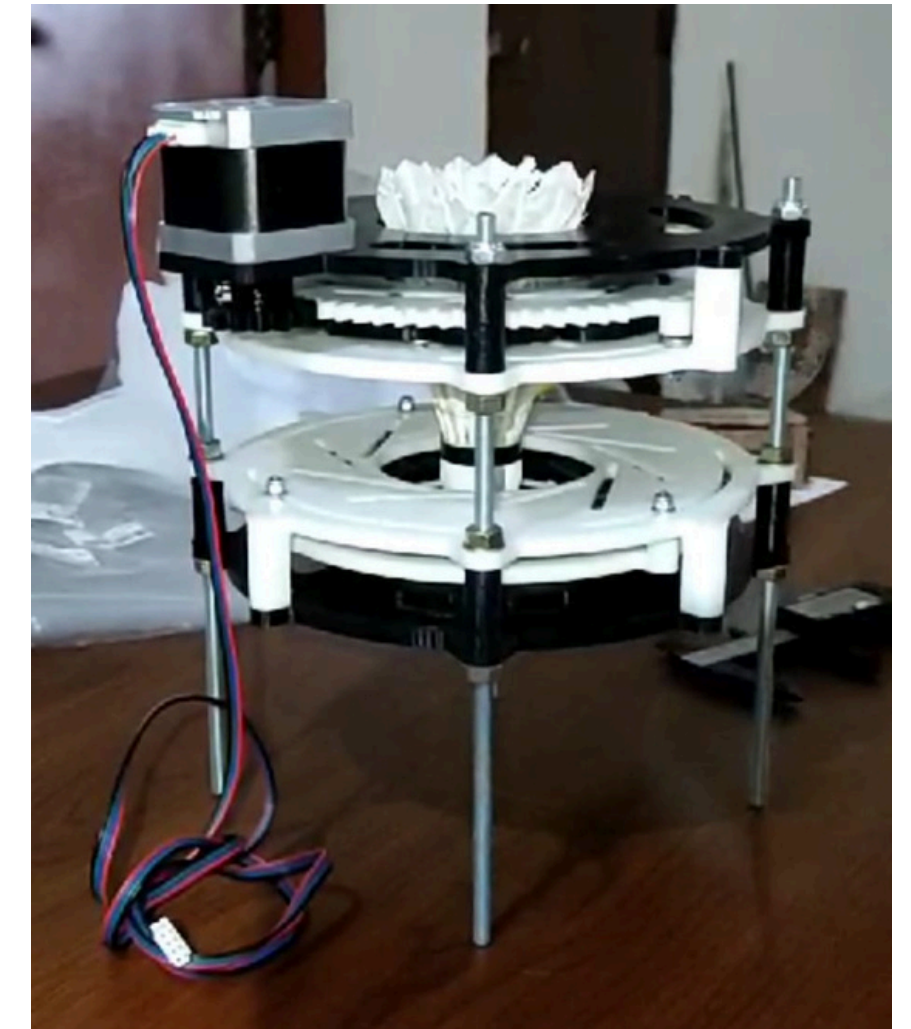


Figure 11: Assembled Prototype

DESIGN AND DEVELOPMENT OF ROBOTICS KITS FOR OPTIMUS ROBOTICS

LANKATRONICS (PVT) LTD.

Overview:

This involved with the design and manufacturing of DIY robotic products under the brand **Optimus Robotics**, a subsidiary of Lankatronics (Pvt) Ltd. This main aim of this brand is to create accessible, user-friendly robotics kits for hobbyists and educators.

Objectives:

1. Develop modular and customizable robotic products that cater to various skill levels
2. Ensure an easy assembly with comprehensive instructions and support material

Prototyping:

3D Printing using PLA+, ABS+, PETG and TPU depending on the functionality

Design Tools:

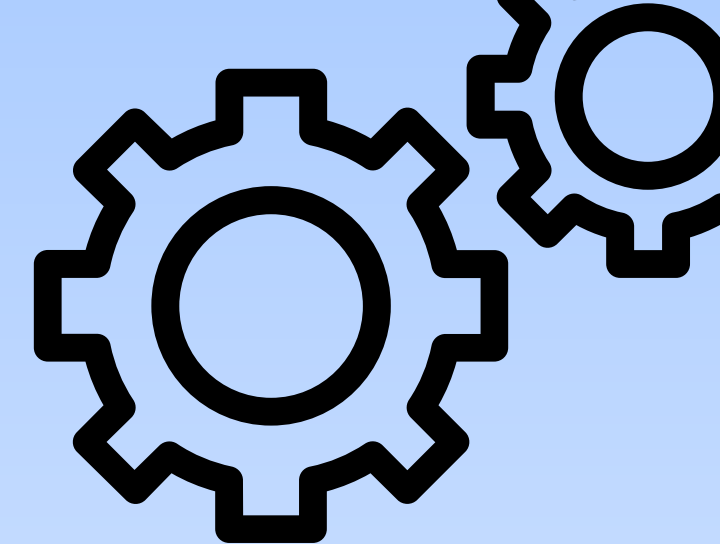
SolidWorks - Solid Modelling and Motion Study

Firmware Development:

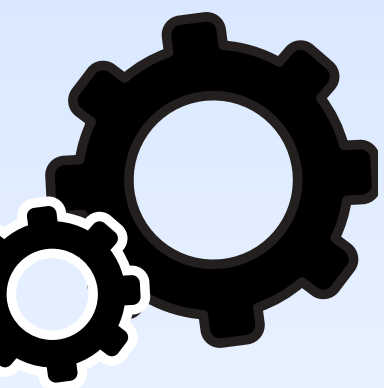
C/C++ using Arduino/VS Code



Figure 12: Samples of Optimus Robotics Kits



FREELANCE PROJECTS



GPS TRACKER ENCLOSURE DESIGN, ASSEMBLY AND MANUFACTURING CONSULTATION

Predictiv AI

Requirements:

1. Optimize enclosure for the provided PCBA allowing necessary clearance i.e. SMA antenna connectors, Power jacks, Mounting Bosses, etc.
2. Provide LED status indication on the enclosure
3. Provide STL, STEP and other necessary manufacture file formats
4. Provide Assembly Guideline/Manual

Prototyping:

1. 3D Printing - For the enclosure using ABS+
2. Laser Cutting - Light Pipe profiles used for status indication

Design Tools:

SolidWorks - Solid Modelling, Assembly, Design for Manufacturing and Assembly

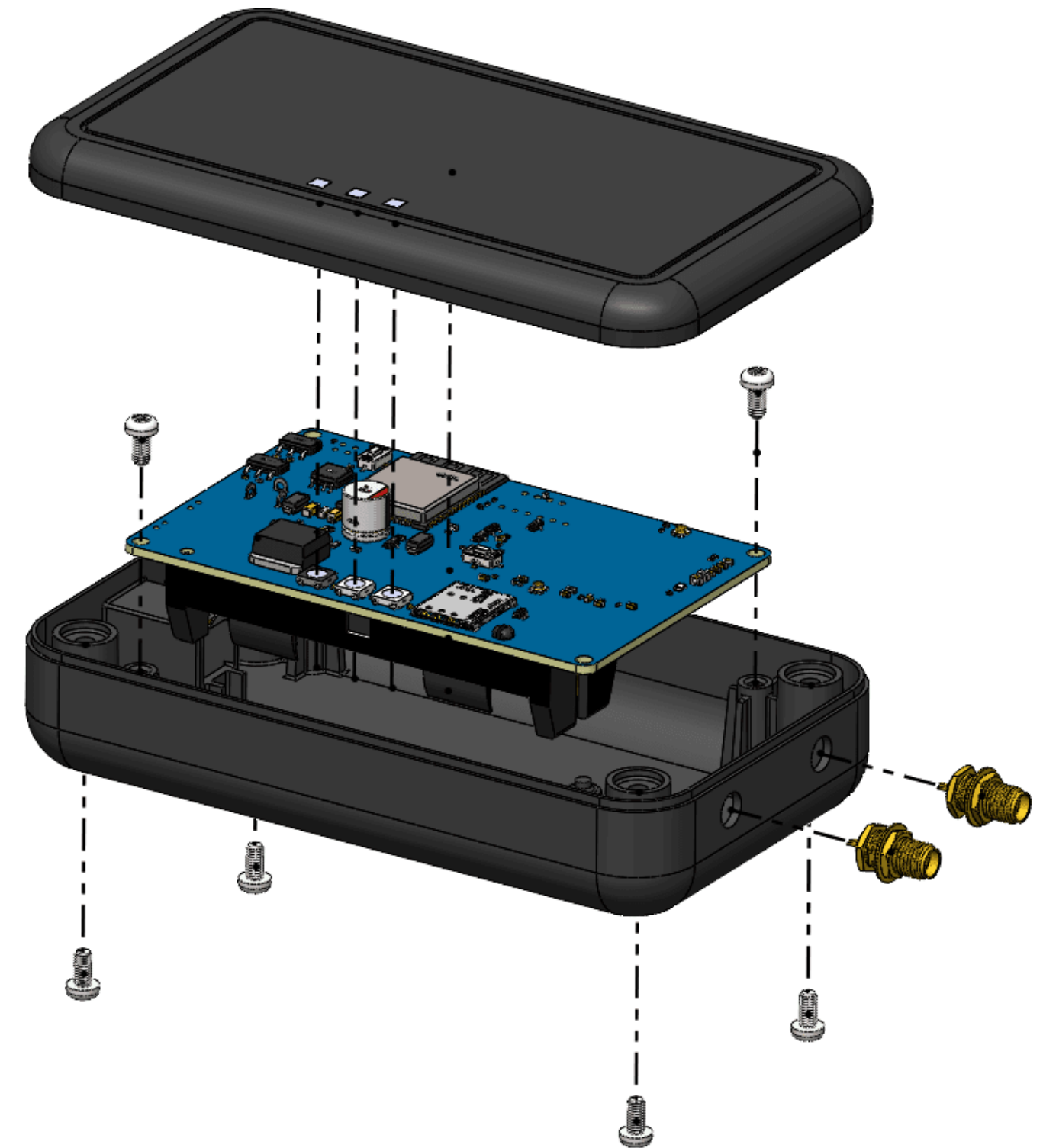


Figure 13: Exploded View of the GPS Tracker Enclosure

OBD2 PCB ENCLOSURE DESIGN, ASSEMBLY AND MANUFACTURING CONSULTATION

Predictiv AI

Requirements:

1. Optimize enclosure for the provided PCBA allowing necessary clearances for Mounting Bosses.
2. Provide STL, STEP and other necessary manufacture file formats
3. Provide Assembly Guideline/Manual

Design Tools:

SolidWorks - Solid Modelling, Assembly, Design for Manufacturing and Assembly

Prototyping:

3D Printing - For the enclosure using ABS+

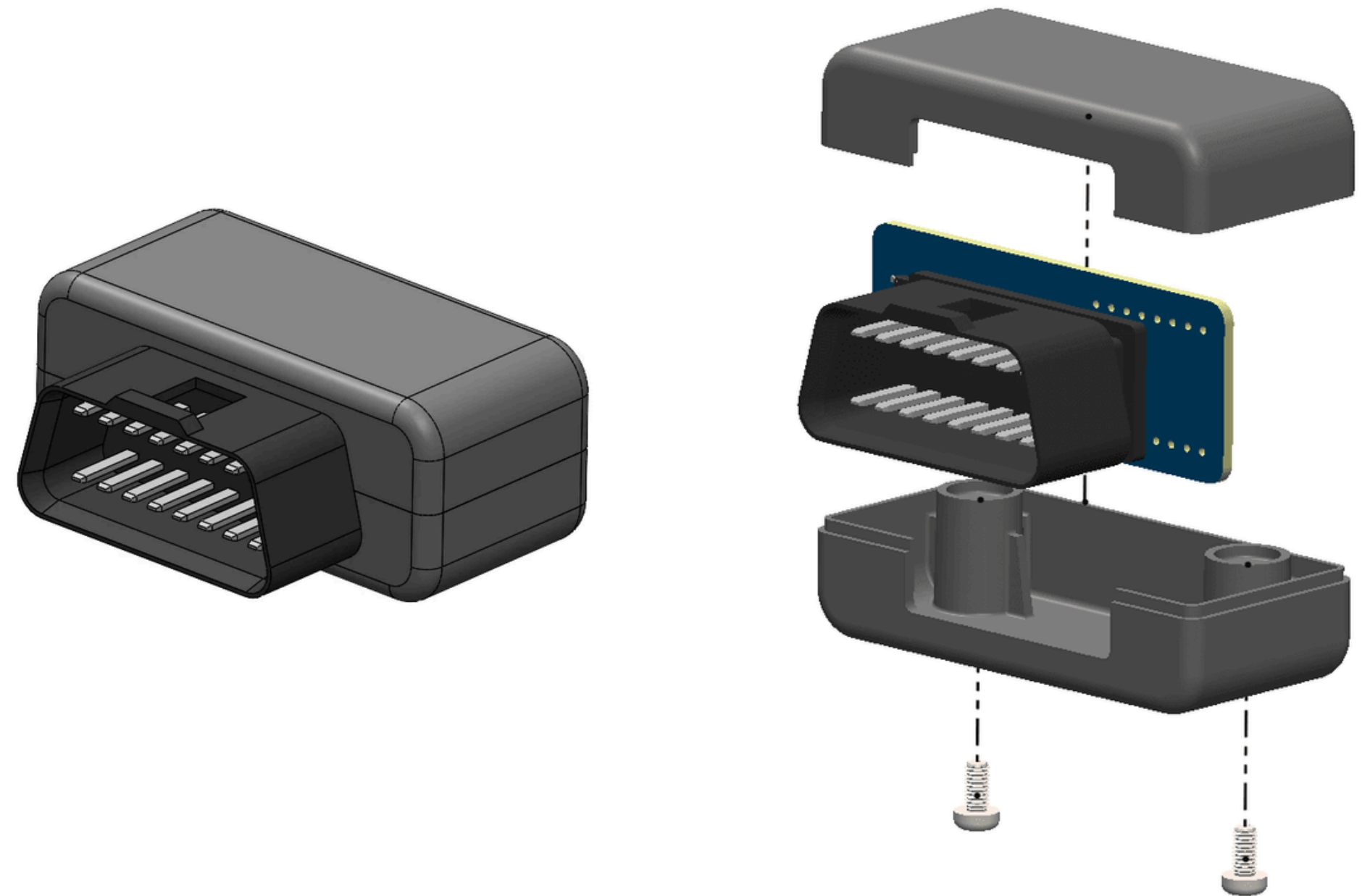
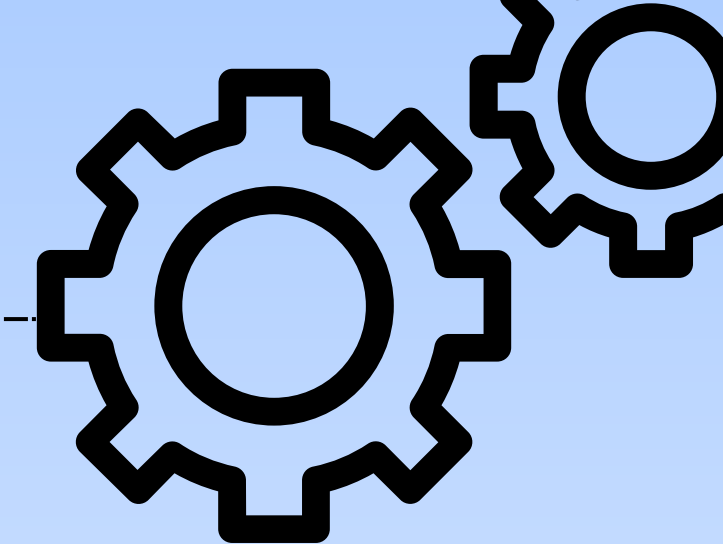


Figure 14: Isometric and Exploded Views of the OBD2 PCB Enclosure



THE END

